

1.1 Basic Addition with Carrying

1. Word Problem: If Ron has 2127 pieces of candy and Harry has 5374 pieces of candy, how much candy do they have in total?

7,501 pieces

2. Solve the following problems.

$$\begin{array}{r} 11 \\ 128 \\ + 284 \\ \hline 412 \end{array}$$

$$\begin{array}{r} 34 \\ + 28 \\ \hline 62 \end{array}$$

$$\begin{array}{r} 64 \\ + 48 \\ \hline 112 \end{array}$$

$$\begin{array}{r} 894 \\ + 689 \\ \hline 1583 \end{array}$$

$$\begin{array}{r} 102 \\ + 9898 \\ \hline 10000 \end{array}$$

$$\begin{array}{r} 1111 \\ + 7898 \\ \hline 9009 \end{array}$$

$$\begin{array}{r} 20102 \\ + 3256 \\ \hline 23358 \end{array}$$

$$\begin{array}{r} 357 \\ + 79898 \\ \hline 80255 \end{array}$$

3. Set up the following problems with one number above the other and then use addition with carrying to find the answers.

$194 + 329 = 523$

$46 + 98 = 144$

$60 + 79 = 139$

$$\begin{array}{r} 11 \\ 194 \\ + 329 \\ \hline 523 \end{array}$$

$987 + 568 = 1555$

$695 + 1234 = 1929$

$1111 + 2568 = 3679$

4. Challenge: Emma added a 3 digit number ABC to 782 using addition with carrying. She carried ones in the places I've shown. What is the largest possible answer that she could have gotten? (Assume that A, B, and C are all different digits.)

$$\begin{array}{r} 1 \quad 1 \\ 782 \\ + ABC \\ \hline \end{array}$$

1690

Notes on 4: To make the sum as large as possible you want A as large as possible. If A=9 is possible that will be the solution, so try A=9. In the tens column $8 + B + 1$ does not result in anything being carried to the hundreds column, so we must have B=0. In the ones column $2 + C$ results in a 1 being carried to the tens column, so we must have C=8 or 9. We used 9 already so use C=8. $782 + 908 = \boxed{1690}$.

1.2 Working with Bigger Numbers

1. Word Problem: Ingrid made 5768425392 Zimbabwean dollars from selling squid and 21736808 Zimbabwean dollars from selling shark. How much money did she make in all?

5,790,162,200 Zimbabwean dollars

2. Insert commas into the following numbers.

4, 7 8 9, 5 6 2 8, 9 5 6, 7 7 6, 2 1 7 7, 5 2 1, 4 8 9, 6 3 2

3. Write the following numbers out in words.

4789562 Four million, seven hundred eighty nine thousand, five hundred sixty two

7521489632 Seven billion, five hundred twenty one million, four hundred eighty nine thousand, six hundred thirty two

4. Arrange the following addition problems so that one number is above the other and solve.

$$4789562 + 5562842 = 10,352,404 \quad 1243217 + 875624 = 2,118,841$$

$$\begin{array}{r} 1111111 \\ 4,789,562 \\ + 5,562,842 \\ \hline 10,352,404 \end{array}$$

5. The Earth's orbit around the Sun is not a perfect circle. When the Earth is closest to the Sun in early January it is about 91,402,640 miles from the Sun. When the Earth is farthest from the Sun in early July it is 3,106,820 miles farther from the Sun. How far from the Sun is the Earth at its farthest point?

94,509,460 miles

6. Challenge: What is two hundred fifty six trillion, five hundred twenty nine million, six hundred twenty nine thousand, five hundred three plus twenty nine quintillion, one hundred eighty six quadrillion, two hundred ninety nine trillion, four hundred seven million, sixty seven thousand, one hundred ninety nine?

29,186,555,000,936,696,702

Notes on 6: Just line up the numbers carefully and do addition with carrying. Be careful to put zeros where needed. The names for each three digit block after the millions (in US English not in the UK) are billion, trillion, quadrillion, and quintillion.

$$\begin{array}{r} 256,000,529,629,503 \\ + 29,186,299,000,407,067,199 \\ \hline \end{array}$$

1.3 Working with Decimals

1. Word Problem: Artemis bought \$1043.78 of cameras to mount around his house and \$228.33 of motion detectors. How much did he spend in total on cameras and motion detectors?

$\$1272.11$

2. Insert commas into the following numbers.

1 3,8 2 8.2 3 2

1, 2 9 5, 8 6 2. 4 8 6 9

7 8 9, 6 5 8. 4 7 8 5

3. Write the following numbers out in words.

13828.232 thirteen thousand, eight hundred twenty eight and two hundred thirty two thousandths

752.15 seven hundred fifty two and fifteen hundredths

5000964.263 five million, nine hundred sixty four and two hundred sixty three thousandths

4. Arrange the following addition problems so that one number is above the other and use addition with carrying to solve.

$$12.48 + 78.15 = 90.63$$

$$124.6 + 35.35 = 159.95$$

$$337.78 + 23.441 = 361.221$$

$$87654321.01234 + 123.45 =$$

$$87,654,444.46234$$

5. Challenge: At Ralphs's Sandwich Shop a turkey sandwich costs \$5.56, a roast beef sandwich costs \$7.72, a chicken salad sandwich costs \$6.21, a prosciutto and mozzarella sandwich costs \$9.34, and a peanut butter and jelly sandwich costs \$3.75 Megan's mother buys three sandwiches that together cost exactly \$21. Which sandwiches did she buy?

Two roast beef sandwiches and one turkey sandwich

Notes on 5: One approach is to focus on trying to get the sum to be an even number of dollars. The solution must involve at least two different sandwiches, so you can organize your work around a list of the ten possible pairs written in some logical order, e.g turkey+rb, turkey+cs, turkey+pm, etc. Then, for each pair add the cents and see if there is a third sandwich that makes a whole number of dollars. The first pair in this order has turkey+rb = 56c + 72c = 28c, so a sandwich with a price ending in 72 will give a whole number of dollars. This suggests the two roast-beef, one turkey combination, which we find does work by adding $5.56 + 7.72 + 7.72$.

1.4 Adding Lists of Numbers with a Pattern

1. Word Problem: Every year, for her birthday, Abigail gets five times her age in dollars. So, for example, when she turned 3, she got \$15, and when she turned 20, she got \$100. If Abigail is 20, how much money has she gotten so far?

\$ 1,050

2. Add the following.

$$\begin{array}{r} 10 \\ 12 \\ 14 \\ +16 \\ \hline 52 \end{array}$$

$$\begin{array}{r} 22 \\ 33 \\ 44 \\ +55 \\ \hline 154 \end{array}$$

$$\begin{array}{r} 798 \\ 8079 \\ 9807 \\ +7980 \\ \hline 26664 \end{array}$$

$$\begin{array}{r} 21 \\ 32 \\ 43 \\ 54 \\ +65 \\ \hline 215 \end{array}$$

$$\begin{array}{r} 100 \\ 101 \\ 110 \\ +111 \\ \hline 422 \end{array}$$

$$\begin{array}{r} 13254 \\ 32541 \\ 25413 \\ 54132 \\ +41325 \\ \hline 166665 \end{array}$$

3. Find the sum of the positive even integers from 30 to 58, i.e. find $30 + 32 + 34 + 36 + \dots + 58$.

660

4. Find the sum of all positive six-digit whole numbers that have one digit equal to 4 and the rest of their digits equal to 1. For example, two such numbers are 111,114 and 141,111.

999,999

5. Challenge: List all three digit numbers that have only 1's and 2's as digits. What is the sum of all of these numbers?

111, 112, 121, 122, 211, 212, 221, 222

The sum is 1,332.

Notes on 5: It is always a good idea to put your lists in a clear, logical order. Here, I organized the numbers from smallest to largest. To add the numbers you can write them as an addition with carrying problem. Each column has 4 1's and 4 2's for a sum of 12. The ones that are carried make the answer 1332.